



Agriculture: Corn Pest Management Guidelines

Head Smut

Sphacelotheca reiliana

Symptoms and Signs

Head smut is characterized by large smut galls that replace [ears\(/PMG/S/D-CN-SREI-GH.003.html\)](#) or tassels. The galls are first covered by fragile, creamy white membranes that eventually rupture to release masses of dark brown spores. Within the masses of spores are more or less intact threadlike strands of vascular bundles, giving the spore masses a stringy appearance. The vascular bundles within the galls readily differentiate head smut from common smut. In head smut, [leaflike proliferations\(/PMG/S/D-CN-SREI-GH.002.html\)](#) often occur in tassels and partially smutted ears.

Comments on the Disease

Spores survive in the soil for long periods (at least several years). The fungus attacks seedlings and the mycelium becomes systemic in apical primordial tissue, invading undifferentiated floral tissues. Ideal conditions for growth are the same as those that favor germination and growth of the corn. Infection level is related to the concentration of spores in the soil. Although the spores may be seedborne, this is not an important source of inoculum. Spores may also be carried in the air during winds, but they drop to the ground and it is there that they germinate and infect the corn.

Management

Use resistant hybrids; most U.S. hybrids are tolerant. Check with your county farm advisors for the best ones to grow in your area. Fungicides are not available for head smut.



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